Certified Rewrite Discovery for Tensor Programs: A Machine-Minted Conv+BN Fold from Minting to Runtime

Draft — proposed, not accepted

catopt project

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Abstract

Compilers choose better implementations only from transformations a human already wrote. We describe a pipeline in which the rewrite rules themselves are a product of measurement: laws are data, composites are minted by a fixpoint over the shipped library, guards are transported through an evidence store, derivability is credited against an object's own premise closure, and admission is decided by a staged gauntlet that has repeatedly falsified the engine's own candidates. The flagship result is `conv_bn_fold` — the eval-mode batch-norm weight fold a hand optimizer writes — minted, admitted, and shipped verbatim as a machine-authored law in the default rule set. The deployed program evaluates bitwise-equal to the spelled one through an independent runtime and is 16% cheaper per call (72.8 vs 87.0 ms), with the term-level cost model agreeing in direction. All reported numbers are reproducible from the committed tree. The scope is deliberately honest: a kernel-scale, certified-rewrite result, not a model-scale speedup claim.

> **!** Important
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> **Status.** This is a working draft — *proposed, not accepted*. It has not had human revision or external review. Every quantitative claim names the probe, test, or retro that produced it; the rest of the paper should be read as a candidate for rewriting, not as a settled result.

1 Introduction

A tensor-program optimizer faces four questions that are usually conflated: *optimization* (choosing a better implementation among known transformations), *discovery* (finding transformations nobody wrote), *verification* (establishing validity before acceptance), and *search* (exploring the space efficiently). Equality saturation (Tate et al. 2009; Willsey et al. 2021) answers the first and third by closing the program space under a fixed law bank and extracting a cheapest member; systems such as TENSAT apply it to tensor graphs with a hand-written rule set. Rule-inference work (Ruler (Nandi et al. 2021), the Babble line (Cao et al. 2023)) attacks discovery for scalar and library-abstraction domains, but the synthesis story for *tensor programs with runtime-priced deployment* is thinner.

This paper reports the machinery — and one complete measured traversal of it — in the catopt engine. The claim is narrow and checkable:

> A rewrite a hand optimizer would write — the eval-mode conv+BN weight fold — was **discovered, proven, priced, and shipped by the machine**, and its deployed form runs bitwise-equal and cheaper through an independent runtime.

No model-scale speedup is claimed. The result is kernel-scale (one op pair, fp64 fixtures), the runtime is a tree-walk interpreter (debug build), and the move alphabet is bounded by an 18-op host-tensor surface — all named in Section 6. What the case study establishes is the *pipeline*: that a law can travel from measured need to a shipped, certified, runtime-priced rule without a human writing it.

The repository is `handl-lang/research-mlopt`; every number below cites the artifact that produced it (§Section 7).

2 Background

2.1 Equality saturation

Given a term and a set of rewrite laws, an e-graph compactly represents the closure: all programs reachable by the laws, sharing subterms, until a fixed point (Tate et al. 2009). Extraction then picks a cheapest member under a cost model. The soundness boundary is the law bank: any unsound rule poisons every e-class it touches, so in catopt a rule's side conditions (`check`/`derive`/declarative `cond`) are the *only* layer that keeps a false member out — the certificate replay re-proves that a step is a valid application, not that the rule is true.

2.2 Laws as data on a categorical substrate

Catopt reads a program as a 1-morphism between tensor types, a rewrite law as a 2-cell between programs, and a derivation — one law proven from premises — as a 3-cell. The consequence that matters operationally: laws are **data rows** (an LHS/RHS pattern pair plus a guard spec), serializable, storable, composable. The shipped bank (`catopt_core.laws.ALL_RULES`, 84 rules) mixes human-authored axioms with twelve machine-admitted objects (`catopt_core.laws.provenance.MACHINE_SHIPPED`).

2.3 The handl substrate and catopt-kit

The same "one language, many interpretations" model is implemented by the handl platform: eval, typecheck and rewrite are interpretations of one language whose tables are data. `packages/catopt-kit` — a plain Rust crate — composes handl's canonical `Language`, its lex/parse/tree-walk-exec engines, and `handl-host-tensor`'s 18-op `tensor.*` surface as dependencies (no forks). `hnl-run` is a subprocess door: `.hnl` in, a JSON wire tensor out, `--repeat N` evaluating a source N times in one process — the measurement hook. `lab/kit/term_hnl.py` renders any catopt IR term as a `.hnl` program from a data table (per-op arg assembly, attr marshaling, leaf binding), so the *same object the e-graph rewrites* is what the runtime executes.

3 Machinery

3.1 The mint fixpoint

`catopt_core.meta.synthesize_rules` runs a bounded fixpoint over the law bank on a seed term: each round applies rules, composes the pairs that chain, and emits `syn_*` composites with provenance — the names of the premise rules each composite was built from. On the eval-mode ConvBN seed it emits 974 rules in 12 s (fuel 30,000, 4 rounds); the fold composite mints deterministically (identical names across runs).

3.2 Guard transport

A minted rule's side conditions are closures over its premise objects, so the evidence store keeps them as *specs*, not code: each stored record carries a `"guard"` field — parent names plus re-expression maps — and `evidence.admit_object` rebuilds `check`/`derive` by resolving parents as store rows, transitively. The shipped form (`catopt_core.laws.minted`) embeds the stored-closure records as data and replays the same rebuild semantics at import — louder than the store: a parent resolving short of declared strength raises rather than admitting a degraded rule.

3.3 Premise-anchored derivability

`verify_law` answers "is this instance derivable under a rule bank?" with a replayable witness. For a stored composite the honest question is *"derivable under the premises it was minted from"* — so the derivability leg evaluates against `base_rules` plus the object's transitive premise closure, with an anti-gaming bar: the certificate must be standalone-replayable and every step naming a non-axiom rule must *site-verify* under the fp64 numeric oracle. A premise false at its applied site revokes the credit and the verdict falls back to the axioms-only answer.

3.4 One cost alphabet

The cost model credits compile-time work structurally: a subtree closed over `Param`/`Const` leaves whose root op is in `FOLD_OPS` — the 161-op fold-at-lowering alphabet, single-sourced between the lowerer, the cost predicate and the extractor — is priced at ~0 marginal. This is what lets a fold's pay leg measure what it should: the fold's `mul`/`reshape`/`rsqrt` chain is once-per-deployment work, not per-call work, and `dag_cost` prices it that way.

3.5 The admission gauntlet

`evidence.run_gauntlet` stages the verdict: reconstruct → full-data → measure → truth → novelty → typed-pay → closure → certificate. Truth is two legs — the numeric oracle over the guarded region *and* premise-anchored derivability; novelty refuses re-spellings of shipped laws (`relation=duplicate`); typed-pay suppresses cost drops that mint ill-typed members; closure bounds the firing footprint. The gauntlet has repeatedly falsified the engine's own candidates — a naive broadcast guard with 212 bad accepts, an apparent paying sibling with semantically unequal sites, four deliberate false twins (`project/retros/mul-unsq-nat.md`, `layer3-loop.md`, `typed-pay-gate.md`).

4 Case study: the conv+BN fold

The rewrite is the standard eval-mode fold:

$$\mathrm{BN}_{\gamma,\beta,\mu,\sigma^2,\varepsilon}(\mathrm{conv}(x,W,b)) \;=\; \mathrm{conv}(x,\; W \odot \hat{s}',\; b \odot s + (\beta - s \odot \mu)), \qquad s = \gamma \cdot (\sigma^2+\varepsilon)^{-1/2}.$$

The account below is the probe's own output (`lab/probes/conv_fold_premise_pay_probe.py`, ~13 s).

Anchor. `CONV_FOLD_RULES` provides the channel-scale and channel-shift anchors (`mul(conv2d(x,W,b),ŝ)`-shaped single-step folds); `BATCH_NORM_RULES` expands `batch_norm` into its affine spelling. Individu-

ally they are enabled-but-opt-in — the anchors refuse at truth-starvation, the expand at typed-pay — their value is *enablement*: they are the fold's derivation premises.

Mint. `synthesize_rules` over the bank minus the already-shipped composite emits 974 rules; the fold composite — `batch_norm(conv2d(·),…)` → `conv2d(·)` — mints as a depth-4 `syn_*` chain over an 8-rule transitive premise closure (five shipped anchors + three intermediates).

Transport and truth. Stored with its closure, `admit_object` rebuilds both hooks from guard records (`missing=('check','derive')` → transported). The transported guard accepts the real site (1 eq / 0 ne) and 105/105 of the guarded synth region with 255 honest declines; `verify_law` replays the derivation under shipped axioms in one step — `witness=('conv_bn_fold',)`, the library law proving its own minted origin.

Pay. On the 8×8 probe site the folded instance drops `dag_cost` **34,874.3 → 17,438.7** (−50 %), fires once, verifies `pass` on the lowered modules.

Ship. The composite is promoted *verbatim*: its stored records embed in `catopt_core.laws.minted`, its guard re-materializes at import, `MACHINE_SHIPPED` grows to 12. Re-running the gauntlet post-ship refuses at novelty with `relation=duplicate` — the documented signature of "already admitted."

Runtime-proven. Through the kit bridge (Section 2), the spelled term `batch_norm(conv2d(x,W,b),…)` and the deployed term `conv2d(x,W2,b2)` — with `W2`,`b2` themselves catopt subterms materialized through the kit — evaluate to the same tensor (**bitwise-equal** on the seeded fixture; the full `torch.export` IR of the module evaluates at 6.7e-16 max abs diff), and per-call wall over `hnl-run --repeat 30` measures:

Table 1: The fold decision, priced by the runtime itself.

arm	per-call (this box)	per-call (recorded)	`dag_cost`
spelled `batch_norm(conv2d)`	87.0 ms	95.4 ms	55,296
deployed `conv2d(x,W2,b2)`	72.8 ms	82.1 ms	36,864
delta	**−16.3 %**	−13.9 %	−33 %

One honest subtlety: `dag_cost`'s param-only discount models compile-time folding the tree-walk runtime does not perform, so the measured arm separates fold-once (materialize `W2`,`b2`) from run-every-call (the deployed conv). The cost model and the wall agree in direction; the wall numbers are debug-build timings — directional, not a benchmark rig.

5 Related work

Equality saturation and e-graph superoptimization. Tate et al. (Tate et al. 2009) introduced the technique; egg (Willsey et al. 2021) made it practical; TENSAT (Yang et al. 2021) and follow-ons apply it to tensor graphs with fixed, hand-written rule sets. Catopt uses the same substrate but the rule bank is not fixed — the discovery loop writes into it through the gauntlet. egglog (Zhang et al. 2023) serves as a test-only differential oracle for the hand-rolled saturation.

Rewrite-rule inference and library learning. Ruler (Nandi et al. 2021) infers rewrite rules for scalar domains by enumeration + equality-saturation filtering — the direct ancestor of the mint-then-verify shape here, differing in that catopt mints *composites over tensor laws* with transported guards and credits derivability against an object's own premise closure. Babble (Cao et al. 2023)

learns library abstractions by anti-unification; catopt's definitional-atom layer is the same move and measured **null** on reach-expansion — the paying layer is above it.

Compilers. TVM (Chen et al. 2018) autotunes schedules; TorchInductor (Ansel et al. 2024) fuses pointwise ops and codegen's the graph a human optimizer could reach. Both are incumbents in the repo's benchmark tables (wins and losses alike); the contribution here is not a faster kernel but a certified, machine-discovered *rewrite*.

Certified/verified compilation. The per-artifact boundary design — law, search, delivery, bound each with its own verifier — is in the spirit of translation validation; what is unusual is that the verifier also audits the *discovery* machinery (adversarial sweeps falsifying minted guards).

6 Limitations

- **Kernel scale.** One op pair, fp64 fixtures, a 1×4×24×24 activation. Nothing here measures model-scale compilation; wider-corpus evidence in the repo (bimodal exposure, ~1.5% yield of fresh paying structure) is honest that this is not yet a general speedup.
- **Move alphabet bound.** The kit surface is 18 `tensor.*` ops; the fold's own `s = ·rsqrt(²+)` is outside it (no elementwise `rsqrt`) and rides a marshaling channel. Emitted terms are confined to what the surface spells; outside-surface ops refuse by name.
- **Tree-walk runtime.** `hnl-run --repeat` timings are wall-clock on a debug-build tree-walk interpreter — they establish direction (the fold's value lands in the deployed program), not competitive throughput. No codegen claim is made.
- **Single-case gauntlet.** The admission measured here is a one-term corpus; the machinery's broader soundness evidence lives in the library-level sweeps, which caught real holes — but on different objects than this one.
- **Attr-narrow LHS.** The shipped composite's LHS carries the seed's literal attrs (`stride=(1,1)`, `padding=(1,1)`, `eps=1e-05`, `training=False`): its region is the default-attr eval-mode fold — honest, narrow, the common spelling.

7 Reproduction

All numbers are reproducible from the committed tree at `handl-lang/research-mlopt`:

```
uv sync
.venv/bin/python lab/probes/conv_fold_premise_pay_probe.py   # the admission account (~13 s)
.venv/bin/python -m pytest tests/test_kit_bridge.py -q       # the runtime-priced decision (8 t
cargo test -p catopt-kit                                     # the Rust consumer (21 tests)
.venv/bin/python -m pytest -m "not slow"                     # the dev lane
```

The evidence ledger is `bench/RESULTS.md`; the measured retros each claim cites are under `project/retros/` (notably `minted-composite-default.md`, `kit-term-bridge.md`, `premise-anchored-derivabi fold-cost-single-source.md`); the extended working draft with the full win/loss tables is `docs/paper_draft.md`.

References

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